KGP-Assist: AI Agent for Issue Resolution at IIT Kharagpur

KGP-Assist is an AI-powered agent designed to address general issues faced by students, faculty, and administrative staff at the Indian Institute of Technology, Kharagpur. This comprehensive system leverages natural language processing, machine learning, and institutional data to provide personalized assistance.

The agent aims to transform the current fragmented issue resolution process into a streamlined, efficient, and user-centric experience. It provides centralized information access, automated issue resolution, personalized assistance, process simplification, 24/7 availability, and data-driven insights.

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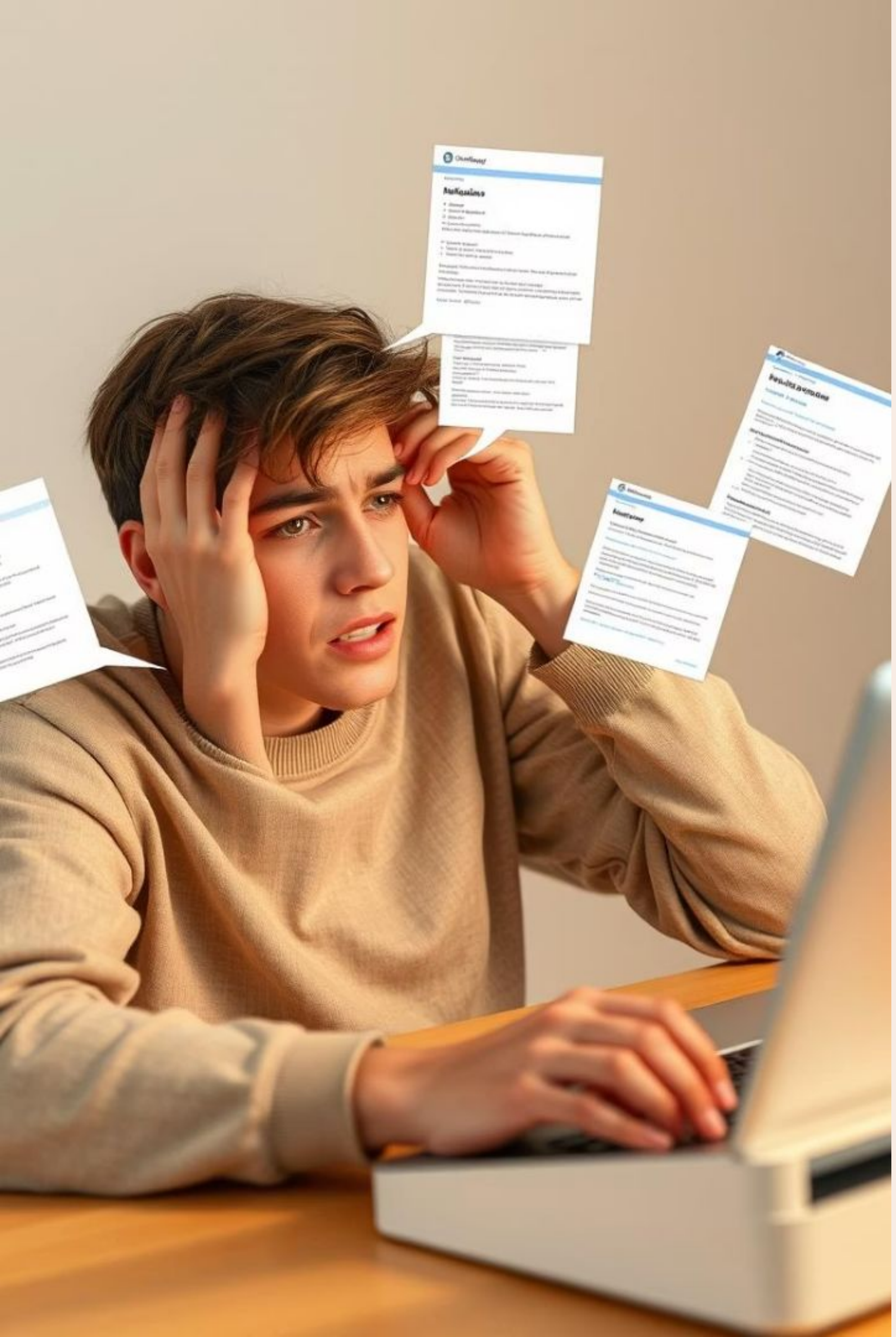


Current Challenges & Problem Statement

Fragmented Information Sources

Critical information is scattered across multiple platforms, departments, and websites, making it difficult for users to find accurate and up-to-date information.

Process Complexity

Many administrative procedures involve multiple steps, approvals, and documentation requirements that are often unclear to users, resulting in repeated queries and submissions.

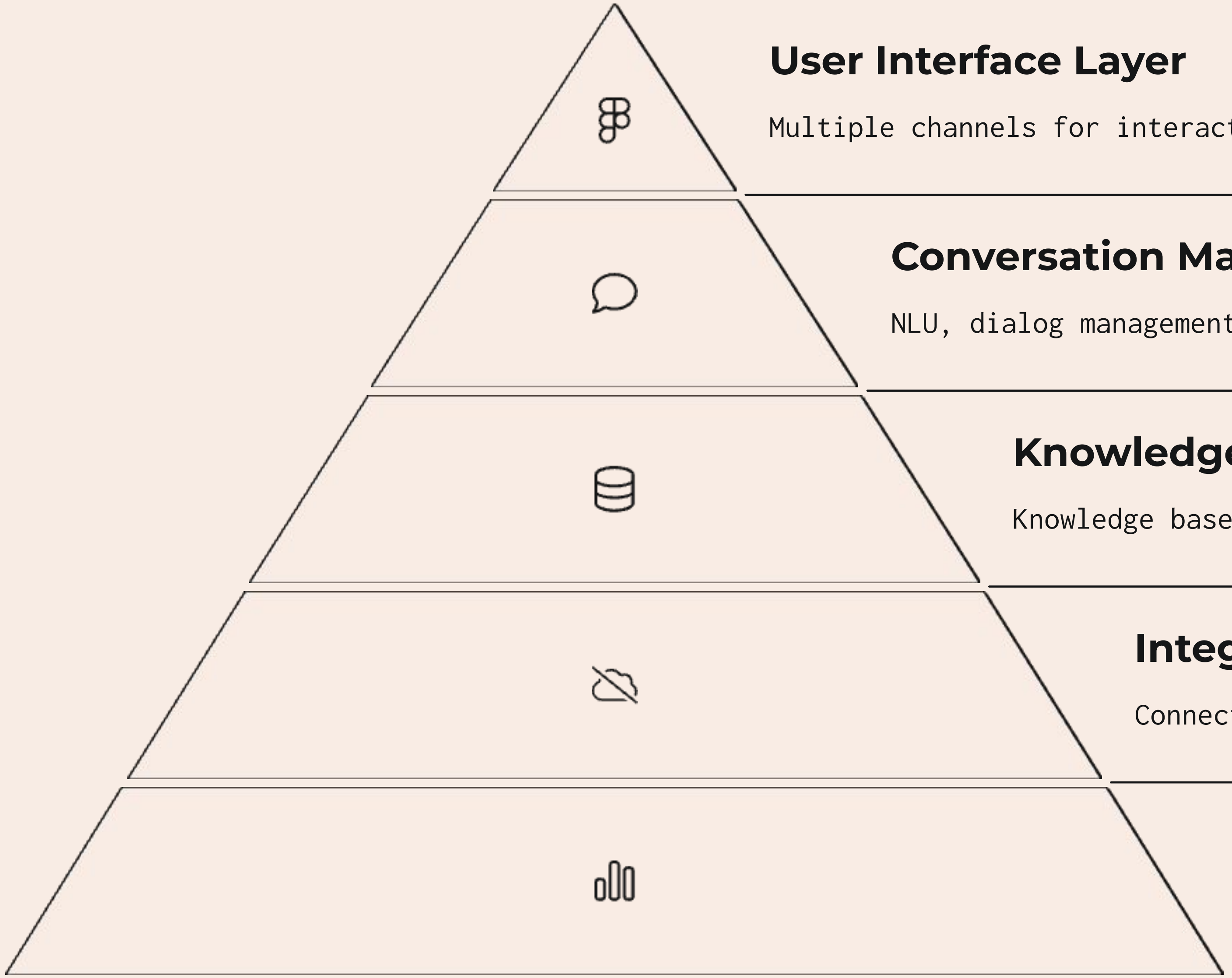
Resource Constraints

Limited administrative staff must handle a high volume of repetitive queries, reducing their capacity to address more complex issues that require human intervention.

Accessibility Barriers

Current support systems primarily operate during business hours, creating accessibility challenges for users who need assistance outside these hours or from remote locations.

System Architecture & Components



User Interface Layer

Multiple channels for interaction

Conversation Management Layer

NLU, dialog management, response generation

Knowledge Processing Layer

Knowledge base, information retrieval, reasoning

Integration Layer

Connects with existing campus systems

Analytics Layer

Collects and analyzes interaction data

The KGP-Assist architecture is designed as a modular, scalable system that integrates multiple components to deliver intelligent issue resolution capabilities. The data flows through user input processing, knowledge retrieval, response generation, action execution, and a feedback loop for continuous improvement.

Key Features & Implementation

Natural Language Understanding

Advanced NLP capabilities to understand and respond to user queries in natural language, supporting both text and voice interactions. Implemented using state-of-the-art language models for intent classification and entity extraction.

Knowledge Base

Comprehensive information repository covering academic regulations, administrative procedures, campus facilities, events, and services. Implemented using a combination of graph database (Neo4j) and document store (Elasticsearch).

Dialog Management

Handles conversation flow, context tracking, and multi-turn interactions. The system determines appropriate actions based on user intent and context, maintaining session history for personalized responses.

System Integration

Connects with existing campus systems through APIs and data synchronization, including academic ERP, facility management, and complaint systems for seamless service delivery.

Evaluation & Future Enhancements

Performance Metrics

User engagement, issue resolution, satisfaction, system performance

Advanced AI Capabilities

Predictive analytics, personalized learning paths, emotion recognition

Extended Integration

Smart campus, research management, alumni network

Expanded User Interfaces

AR interface, smart speakers, digital signage, wearables

The success of KGP-Assist will be measured through comprehensive metrics including user adoption (70% target), issue resolution rate (80% target), user satisfaction (4.5/5 CSAT), operational efficiency (30% workload reduction), and actionable insights generation. Future enhancements will focus on expanding AI capabilities, integration with additional systems, and developing new user interfaces.